

ELEC3705 Fundamentals of Quantum Engineering

Laboratory 2 : Matrix Formalism of Quantum Systems

Abstract

Historically a Stern-Gerlach apparatus was used to determine that silver nuclei had an intrinsic and *quantised* magnetic moment. This lab will utilise this apparatus as a framework to explore how Python can be used to perform measurements on a quantum system described by linear algebra. Measurements of this kind are sometimes referred to as Stern-Gerlach analysis (SGA).

Introduction

In 1922 Otto Stern and Walther Gerlach designed an experiment which revealed that the magnetic moment is an intrinsic quantity of nuclei (originally, silver was used), and that it does not arise from the orbital angular momentum of the atom's electrons. Furthermore, this magnetic moment, or 'spin', is quantised, meaning it can only take discrete values. The value of the magnetic moment was measured to be $S_z = \pm \frac{\hbar}{2}$. This is a profound discovery, as one might expect that measuring a three-dimensional quantity in one orientation would yield a continuum of values. What we will explore in this lab is how a quantum measurement will project the state of a quantum system into an eigenstate of the measurement basis.

Stern-Gerlach Experiment

Figure 2 depicts a modified Stern-Gerlach apparatus which at the output will either block or transmit the incident nucleus depending on its spin orientation. Presume that we can detect any nucleus that passes through the slot on the filter plate. If we have a nucleus with randomised spin entering the apparatus, we would expect that the result would be 50% spin-up (z_+) and 50% spin-down (z_-). We indicate this by saying that the expectation value, $\langle S_z \rangle = 0.5 \cdot \frac{\hbar}{2} + 0.5 \cdot \frac{-\hbar}{2} = 0$. Let's now define our measurement basis along the z -axis.

Let

$$|+z\rangle = |\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-z\rangle = |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (1)$$

These will be our two observable states, defined by the orientation of the magnetic field of the Stern-Gerlach apparatus. After measurement we *must* find the spin either in $|\uparrow\rangle$ or $|\downarrow\rangle$. Our

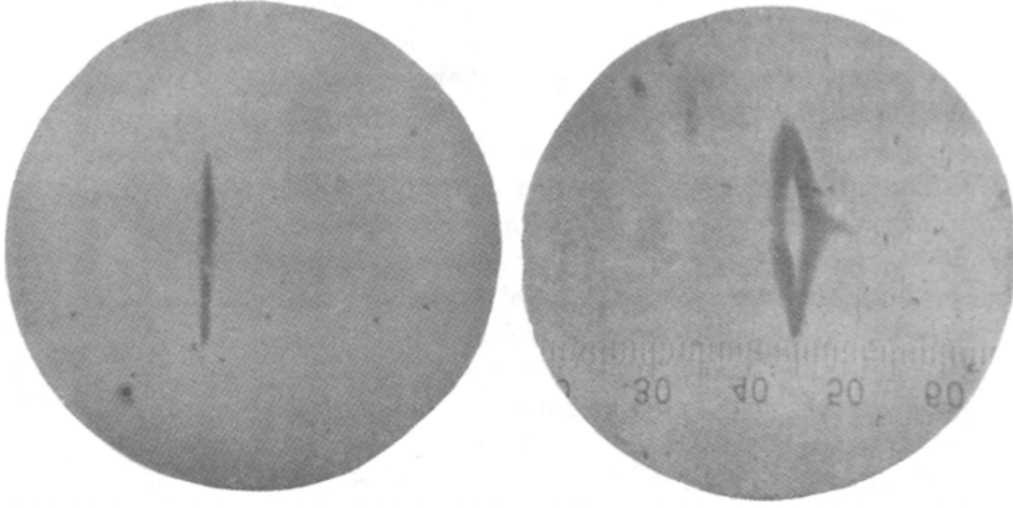


Figure 1: The results of a Stern-Gerlach experiment without an applied external magnetic field (left) and with one (right). Note the discrete 2 level splitting of the incident particles, which are assumed to have random 3D orientation when entering the apparatus.

measurement basis will be composed from the Pauli spin matrices (Equation 2), which will be utilised extensively in the following labs. For now, we will just use them for measurement calculations. Note that the factor of $\frac{\hbar}{2}$ is ignored for simplicity, so a measurement outcome will now always be ± 1 , i.e. the eigenvalues of each matrix.

$$\left\{ \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right\} \quad (2)$$

Measurement is a statistical affair, where we can determine the means and variances of each measurement, but never deterministically detail what a single outcome will be. The mean (average) value of a measurement can be found by

$$\langle A \rangle_\psi = \langle \psi | A | \psi \rangle \quad (3)$$

where A is an ‘observable’ of your quantum mechanical system and $|\psi\rangle$ the state of the system. Also recall that $\langle \psi | = |\psi\rangle^\dagger$, where $\dagger$ represents the complex conjugate transpose. Further, the probability of getting the result a_n from a measurement is given by

$$P(a_n) = |\langle u_n | \psi \rangle|^2 \quad (4)$$

where a_n is the eigenvalue corresponding to the eigenvector, u_n , of your observable A . To make this more concrete, consider a measurement in the $\vec{z}$ -axis, i.e. $A = \sigma_z$,

$$\langle \sigma_z \rangle_\psi = \langle \psi | \sigma_z | \psi \rangle \quad (5)$$

If we let $|\psi\rangle = |\uparrow\rangle$, we find that the expectation value is 1, and conversely -1 for when $|\psi\rangle = |\downarrow\rangle$.

Question 1

What is the expectation value when measuring the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ in the $\vec{x}$ -axis? Note that measuring along an axis can be interpreted as sending the spin through a Stern-Gerlach apparatus with a magnetic field along said axis. Repeat this question for measurement in the $\vec{y}$ -axis and the $\vec{z}$ -axis. What does the state vector $|\psi\rangle$ represent, i.e. in what direction is spin oriented?

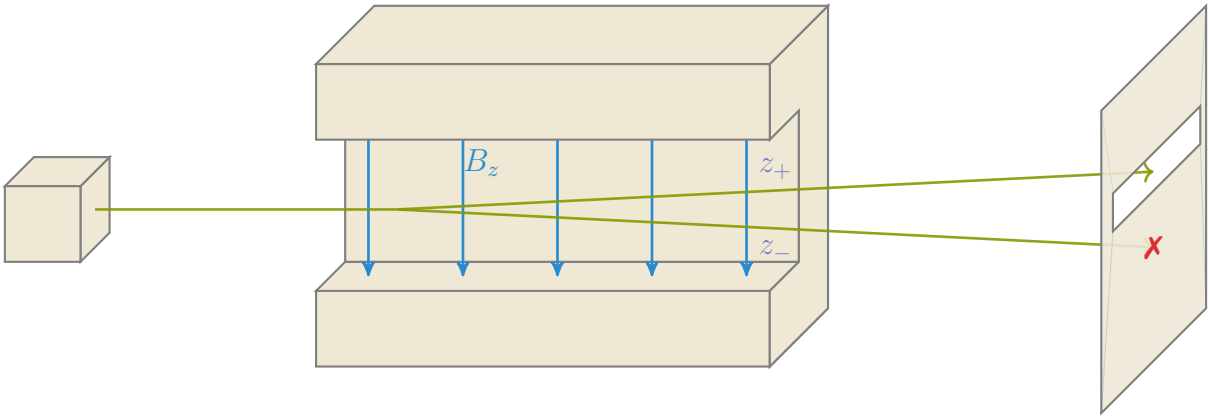


Figure 2: A Stern-Gerlach Apparatus

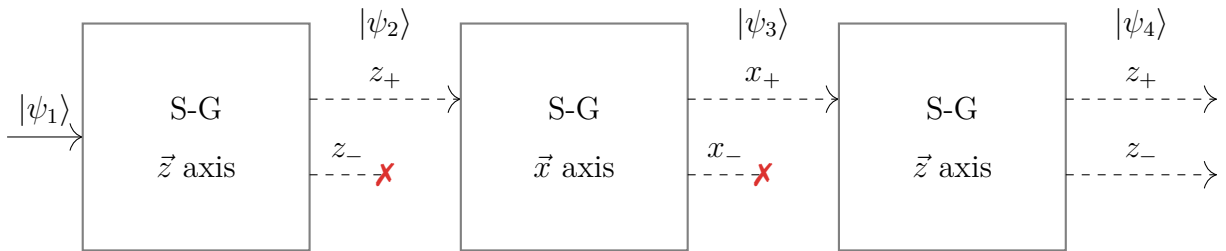


Figure 3: A Stern-Gerlach sequence

The Stern-Gerlach apparatus can be chained together as shown in Figure 3. It is important to reiterate that the state vector entering the second apparatus, $|\psi_2\rangle = |\uparrow\rangle$, will then be projected into either x_+ or x_- . This is a useful tool for simple quantum measurements. Note that you are not limited to measuring in the 3 axes, $\vec{x}$, $\vec{y}$ and $\vec{z}$, but can construct an arbitrary axis of measurement from the Pauli matrices and a unit vector, $\vec{u}$. θ is the angle measured from the positive $\vec{z}$ -axis and ϕ is the angle measured from the positive $\vec{x}$ -axis, counter-clockwise.

$$\begin{aligned}\sigma_u &= (\vec{x} \cdot \vec{u}) \sigma_x + (\vec{y} \cdot \vec{u}) \sigma_y + (\vec{z} \cdot \vec{u}) \sigma_z \\&= \sin(\theta) \cos(\phi) \sigma_x + \sin(\theta) \sin(\phi) \sigma_y + \cos(\theta) \sigma_z \\&= \begin{pmatrix} \cos(\theta) & \sin(\theta)(\cos(\phi) - i \sin(\phi)) \\ \sin(\theta)(\cos(\phi) + i \sin(\phi)) & -\cos(\theta) \end{pmatrix} \\&= \begin{pmatrix} \cos(\theta) & \sin(\theta) e^{-i\phi} \\ \sin(\theta) e^{i\phi} & -\cos(\theta) \end{pmatrix}\end{aligned}\tag{6}$$

Question 2

What are the possible state vectors that $|\psi\rangle$ will be projected into when measuring with a Stern-Gerlach $\vec{x}$ -axis apparatus?

Hint: The eigenvectors of σ_z are $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Question 3

Given the apparatus sequence shown in Figure 3 and a randomised starting state $|\psi\rangle$, i.e. the probability of making it through the first S-G apparatus is 50%, what is the probability that it will be output in the final z_+ state?

Task

Create a Jupyter notebook script which can simulate *any* sequence of Stern-Gerlach apparatus which each perform a measurement along an arbitrary 3-dimensional axis. Your first step will be to create a general Stern-Gerlach apparatus, parametrised by θ and ϕ , and then to propagate an initial state $|\psi_1\rangle$ through the S-G sequence, modifying the output state vector of each apparatus appropriately, i.e. calculating $|\psi_2\rangle$, $|\psi_3\rangle$, etc. You must calculate the expectation value at each point in the sequence, as well as the cumulative probability of the particle arriving at the output of each apparatus. Presume that only the state with eigenvalue 1 will be transmitted to the next stage, as in Figure 3.

a) $|\psi_1\rangle = |\uparrow\rangle$

	X	X	Z
$\langle A \rangle_\psi$			
$P(-1)$			
$P(+1)$			

b) $|\psi_1\rangle = \frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1+i}{2}|\downarrow\rangle$

	X	X	Z
$\langle A \rangle_\psi$			
$P(-1)$			
$P(+1)$			

c) $|\psi_1\rangle = |\downarrow\rangle$

	X	U ($\theta = 35^\circ, \phi = 20^\circ$)	Z
$\langle A \rangle_\psi$			
$P(-1)$			
$P(+1)$			

Extended Task

Simulate a contiguous set of n Stern-Gerlach apparatus where each successive apparatus is slightly rotated, where the final apparatus is rotated by 180° compared to the first. The first apparatus should be aligned with $+z$, and the final apparatus with $-z$, and the initial state

vector is $|\psi\rangle = |\uparrow\rangle$. What is the probability of transmission through the full sequence for varying n ? Note that the angle ϕ in the $x - y$ plane should be kept constant at 0.